

Logic Design Coursework

Problem Set: Truth Tables, Logic Gates, K-Maps, and Adders

Part 1: Truth Tables & Boolean Simplification

Problem 1

Given the following Boolean function: $F(A, B, C) = \sum m(0, 2, 4, 6)$

1. Construct the Truth Table for this function.
2. Derive the unsimplified Sum of Products (SOP) expression.
3. Use Boolean algebra rules to simplify the expression to its minimal form.

Problem 2

Given the Boolean expression: $Y = (A \cdot B') + (A \cdot B) + (A' \cdot B)$

1. Draw the Truth Table corresponding to this expression.
2. Prove mathematically (using Boolean algebra theorems) that the final simplified equation is $Y = A + B$.

Part 2: Logic Gates & Circuit Design

Problem 3

Draw the logic circuit that represents the following Boolean expression using basic logic gates (AND, OR, NOT):

$$F = (X + Y') \cdot (W' + Z) \cdot X$$

Problem 4

A facility requires an alarm circuit that triggers an output ($F = 1$) based on three sensors (A, B, and C). The required condition is: "Sensor A must be active, AND simultaneously, either Sensor B is inactive OR Sensor C is active."

1. Write the Boolean expression for this circuit.

2. Draw the corresponding logic circuit using the appropriate logic gates.

Part 3: Karnaugh Maps (K-Maps)

Problem 5

Using a 3-variable Karnaugh Map (X, Y, Z), simplify the following Boolean function to obtain an expression that requires the minimum number of logic gates:

$$F(X, Y, Z) = \sum m(1, 3, 5, 7)$$

Problem 6

Given the following 4-variable Boolean function:

$$F(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

1. Draw the corresponding 4-variable K-Map.
2. Identify and clearly group the 1s optimally.
3. Derive the final simplified Sum of Products (SOP) expression.

Part 4: The Half Adder

Problem 7

Design a standard Half Adder circuit from scratch:

1. Write the Truth Table for adding two single-bit binary inputs (A and B).
2. Extract the Boolean equations for both the Sum and the Carry.
3. Draw the logic circuit using one XOR gate and one AND gate.

Problem 8

If the XOR gate in a standard Half Adder is strictly replaced by NAND gates (acting as universal gates), draw the new logic diagram, clearly demonstrating how the Sum signal is generated.

Part 5: The Full Adder

Problem 9

A Full Adder is designed to add three input bits (A, B, and Cin).

1. Construct the Truth Table for the Full Adder.
2. Derive the Boolean expression for the final carry-out (Cout) and simplify it to reach the standard form:

$$C_{out} = (A \cdot B) + C_{in} \cdot (A \oplus B)$$

Problem 10

Using a block diagram approach, design a Full Adder utilizing exactly "Two Half Adders" and one OR gate. Trace the internal signals and clearly demonstrate how the carry signals from both Half Adders are combined.